1. **List out the activities to carry out IEE and EIA studies. How do you access the sediment load of a river?**

Answer:

The activities to carry out IEE and EIA studies are:

Phase 1: Pre-Study & Scoping Activities

i. **Determination of the Study Level:** The proponent determines whether the project requires an IEE or a full EIA based on the technical, physical, and social thresholds prescribed in the schedules of the Regulations.

ii. **Scoping (Mandatory for EIA):**

   a. Publish a 7-day public notice in a national daily newspaper to solicit written comments and suggestions from local communities, schools, hospitals, and local bodies.

   b. Draft the Scoping Document based on public input and submit it to the Concerned Ministry for official approval.

iii. **Preparation and Approval of the Terms of Reference (ToR / Work Schedule):**

   a. Formulate a detailed Work Schedule/ToR outlining the study scope, methodologies, expert team, budget, and timeline.

   b. Submit the ToR to the Concerned Body/Ministry for review and official approval prior to starting the field study.

Phase 2: Baseline Data Collection & Field Investigations

i. **Field Surveys and Environmental Sampling:** Collect primary and secondary data within the project-affected area across three major domains:

   a. **Physical Environment:** Air quality, noise levels, geology, soil, hydrology, and water quality.

   b. **Biological Environment:** Assessment of forests, vegetation, wildlife habitats, rare/endangered species, and aquatic biodiversity.

   c. **Socio-economic and Cultural Environment:** Demography, local livelihoods, public infrastructure, and proximity to national, historical, or religious heritage sites.

ii. **Detailed Analysis of Alternatives:** Evaluate various alternatives for the project (such as alternative alignments, designs, technologies, and construction materials) to identify the most sustainable, implementable option with the lowest environmental footprint.

Phase 3: Public Participation & Local Consent

i. **Pasting of Public Notices (Muchulka):** Paste a public notice in the project-affected Rural Municipality/Municipality, local schools, and hospitals to notify stakeholders. A formal deed of public inquiry (**Muchulka**) must be signed by local witnesses to legally certify the notice was posted.

ii. **Conducting a Public Hearing:** Organize a formal **Public Hearing** (सार्वजनिक सुनुवाइ) in the project-affected area. Proponents must present the project details, record local comments, and document the event with attendance sheets, photographs, and audio/visual recordings.

iii. **Obtaining Local Recommendation Letters:** Apply to the concerned Local Level (Rural Municipality/Municipality) to secure an official **Recommendation Letter** confirming that the local government consents to the project under the condition that the environmental mitigation measures are executed.

Phase 4: Impact Assessment, EMP Formulation, & Report Writing

i. **Impact Identification and Prediction:** Systematically identify and evaluate potential positive and negative, short-term, mid-term, and long-term impacts of the project on the environment, public health, and local communities.

ii. **Formulating the Environmental Management Plan (EMP):**
   a. Detail specific **Preventive, Corrective, and Compensatory mitigation measures** for every identified adverse impact.
   b. Incorporate an action plan that outlines monitoring indicators, schedules, budgets, and defines institutional responsibilities.

iii. **Drafting the Final Report:** Write the environmental study report in the prescribed format (Nepali Unicode font size 12, or English Times New Roman font size 12 if there is foreign investment).

Phase 5: Submission, Review, and Approval

i. **Submission for Approval:** Submit the finalized environmental study report, approved ToR, Muchulkas, Public Hearing documents, and Local Recommendation Letters to the Concerned Body (for IEE) or Ministry of Forests and Environment (for EIA).

ii. **Inquiry and Presentation:** Present the study findings before a technical review committee consisting of subject-matter experts and government representatives.

iii. **Official Approval:** Address any queries or revisions requested by the committee. Once satisfied that the proposal will not cause unmitigable significant adverse impacts, the concerned authority issues an official Approval Letter outlining binding terms and conditions for project execution.

Sediment load is the quantity of sediment transported by a river in a given time. It is generally classified into bed load, suspended load and wash load. For hydropower and reservoir projects in Nepal, sediment assessment is important because Himalayan rivers carry large quantities of sediment, particularly during the monsoon. It affects reservoir storage, turbine abrasion, intake operation, desanding-basin design and the useful life of the project.

Methods of assessing sediment load

i. **Suspended sediment measurement**

Water samples are collected at different points across the river section and at different depths using sediment samplers. The samples are filtered, dried and weighed to determine sediment concentration.

$$If \quad C_s is \; sediment \; concentration \; \wedge \; Q \; is \; river \; discharge:$$

$$Q_s = C_s Q$$

$$where \; Q_s \; is \; suspended \; sediment \; discharge.$$

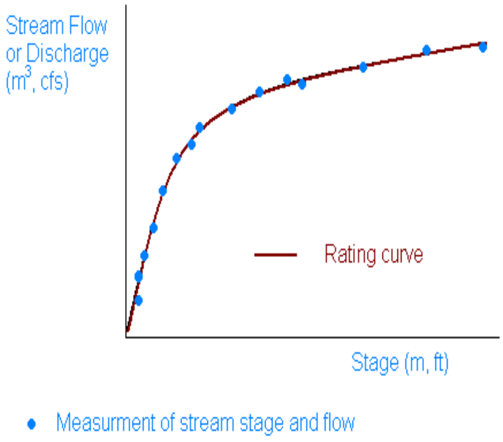
$$For \; concentration \in \mathrm{kg/m^3}, Q_s \; is \; obtained \in \mathrm{kg/s}.$$

ii. **Sediment rating curve**

Repeated measurements of discharge and sediment concentration are used to establish a relationship between water discharge and sediment discharge:

$$Q_s = a Q^b$$

$$where \; a \wedge b \; are \; empirical \; constants.$$



This relationship can then be used with long-term river-flow records to estimate the annual sediment load.

iii. **Reservoir sedimentation survey**

For an existing storage reservoir, the actual sediment deposited in the reservoir can be determined by bathymetric/topographic surveys. Comparing the original and present reservoir capacity gives the amount of sediment deposited.

Total sediment load

The total sediment load can approximately be expressed as:

$$Q_T = Q_{SL} + Q_{BL}$$

$$where:$$

$$Q_T = total \quad sediment \quad load$$

$$Q_{SL} = suspended\text{-}load \quad discharge$$

$$Q_{BL} = bed\text{-}load \quad discharge$$

2. **What is multipurpose water resource development in Nepalese context? Highlight the concept of screening and ranking of hydropower projects.**

Answer:

Multipurpose water resource development means the planned and integrated development of available water resources to satisfy several water-related needs simultaneously, rather than developing water for only one purpose. In Nepal, where rivers have high hydropower potential but water availability varies greatly between the monsoon and dry seasons, multipurpose development is particularly important. The Government's water-resource planning approach promotes integrated use for sectors including hydropower, irrigation, drinking water, flood management, industrial use and navigation.

Nepal receives a large proportion of its annual precipitation during the monsoon, while many areas experience dry-season water shortages. Storage and inter-basin projects can therefore regulate monsoon water and make it available when required. Multipurpose development allows the same dam, reservoir, diversion structure and associated infrastructure to provide several benefits, thereby reducing the unit cost of individual services. Nepal's National Water Plan explicitly notes that multipurpose projects can minimize the unit cost of services and identifies Sunkoshi–Kamala Diversion, West Rapti, Kankai and Bheri–Babai Diversion as multipurpose projects.

For example, in the Bheri–Babai Diversion project, water diverted from the Bheri River supports irrigation development in the Babai area while the project also provides hydropower benefits. The National Water Plan treated the common dam cost as shared between irrigation and hydropower in its economic analysis.

Multipurpose development should be based on river-basin planning, rather than developing individual projects independently. This means considering the upstream and downstream effects, competing water demands, environmental requirements, sedimentation, flood risks and the equitable sharing of benefits. Nepal's Water and Energy Commission (WEC) has a specific mandate to review multipurpose, mega and medium-scale water-resource projects and provide recommendations for their implementation.

Screening of hydropower projects is the preliminary process of examining a large number of potential projects and eliminating those that are technically, economically, environmentally, or socially unsuitable. It helps identify feasible projects for further detailed investigation.

Ranking of hydropower projects is the process of comparing the screened feasible projects using selected criteria and arranging them in order of priority. The criteria may include:

- Installed capacity and energy generation
- Construction cost
- Economic benefits and financial viability
- Hydrological potential
- Environmental and social impacts
- Accessibility and infrastructure requirements
- Construction difficulty and geological conditions
- Contribution to national energy demand
- Project implementation time

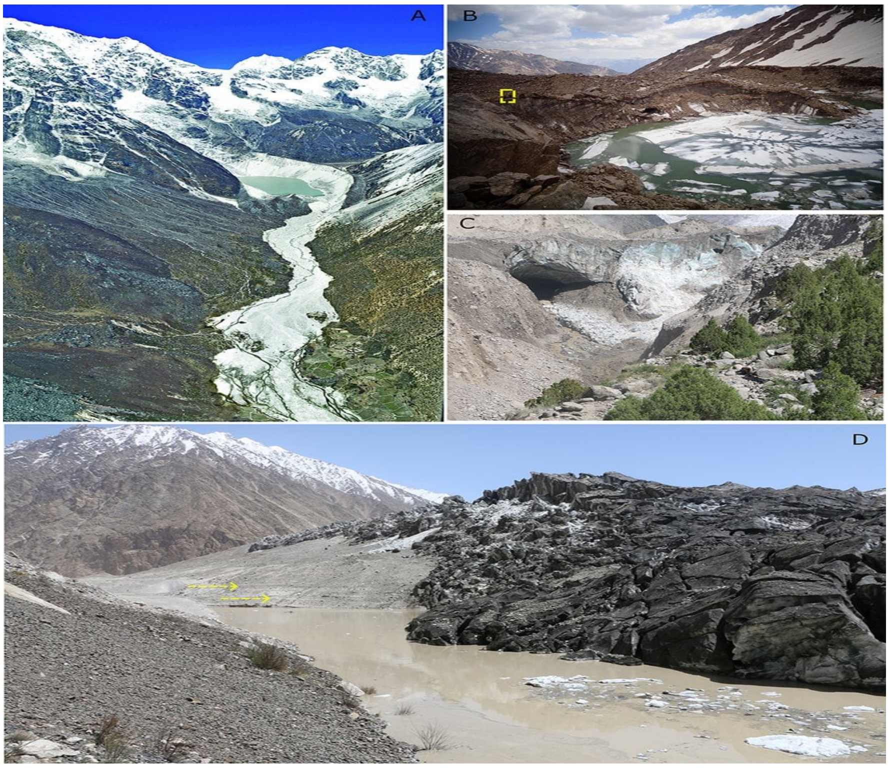
Thus, screening identifies suitable projects, whereas ranking determines their priority for development.

3. **What is GLOF? How do you consider the GLOF phenomenon or event while designing the storage and RoR projects?**

Answer:

GLOF (Glacial Lake Outburst Flood) is a sudden release of water from a glacial lake, usually caused by the failure or overtopping of its natural moraine/ice dam. The resulting flood can travel rapidly downstream carrying water, ice, boulders, sediment and debris, producing very high flood discharges. GLOF is particularly important in Nepal because many major Himalayan rivers originate in glacierized high mountains, and hydropower infrastructure is often located downstream in narrow, steep valleys.

A famous example of GLOF in Nepal is the Dig Tsho GLOF that occurred in 4 August 1985 in the Khumbu region. It destroyed the newly constructed Thame hydropower station, about 30 houses and 14 bridges downstream.



Tsho Rolpa glacial lake located in the Rolwaling Valley of Dolakha is also regarded as one the most dangerous glacial lakes of Nepal. Engineering measures were undertaken around 2000 to lower the lake level and reduce the immediate risk, but the it still requires continuous monitoring as it possess the high risk of GLOF.

The following procedure is generally followed for both storage and RoR projects:

i. **Identify potentially dangerous glacial lakes**
The upstream catchment is investigated using topographic surveys, satellite images, GIS/remote sensing, historical records and field investigations. The size, water volume, moraine condition, lake level and possible triggering mechanisms are assessed.
ii. **Assess the possible GLOF scenario**
Possible failure of the moraine/ice dam is analyzed and the probable breach size, breach formation time, peak discharge, flood volume and flood hydrograph are estimated.
iii. **Route the GLOF downstream**
The GLOF hydrograph is routed from the glacial lake to the proposed project site. The analysis should consider attenuation of the flood as well as sediment, boulders, ice and debris transport.
iv. **Determine the design flood condition**
The GLOF discharge is compared with the normal design flood and extreme flood conditions. The more critical applicable condition is considered for the design of project components.
v. **Check hydraulic structures**
The following structures are checked against GLOF-induced high velocity, scour, debris and flood levels:
    a. Dam/weir and spillway
    b. Intake and approach channel
    c. Undersluice
    d. Stilling basin and apron
    e. River-training works
    f. Headrace canal/tunnel
    g. Penstock
    h. Tailrace
    i. Powerhouse and associated structures
vi. **Consider sediment and debris loading**
GLOF can carry an exceptionally large quantity of sediment and debris. Therefore, the possibility of

intake blockage, abrasion, excessive sedimentation, scour and damage to hydraulic structures must be considered. This is particularly important for Himalayan projects.

vii. **Protect or relocate the powerhouse**
The powerhouse should preferably be located outside the predicted GLOF inundation zone. If this is not possible, suitable river-training, flood-protection and structural protection works should be provided. DoED's Powerhouse Design Guidelines specifically state that the powerhouse should be placed safely after assessment of flood, GLOF and river-damming hazards.

viii. **Provide adequate freeboard and flood protection**
For storage projects, reservoir level rises due to an incoming GLOF, wave action and possible debris blockage should be assessed. Dam and spillway safety must be checked for the resulting extreme condition.

ix. **Provide early-warning and emergency arrangements**
Where significant GLOF risk exists, an early-warning system, communication system and emergency evacuation plan should be incorporated.

4. **What do you mean by hydraulic transient? How do you consider such phenomenon while designing a hydropower plant? A penstock pipe of internal diameter as 2 m is subjected to 100 m statistical head of water and 20% additional dynamic head of allowable stress of material is 120 N/mm^2. Determine the wall thickness of pipe assuming the joint efficiency as 90%.**

Answer:

A hydraulic transient is a temporary change in pressure, flow velocity and discharge in a water-conveyance system caused by a sudden change in flow conditions, such as rapid opening or closing of a turbine/gate.

The most important hydraulic transient in a hydropower plant is water hammer. Rapid reduction in flow produces a pressure rise, while rapid increase in flow can produce a pressure drop.

The approximate water-hammer pressure rise is:

$$\Delta P = \rho a \Delta V$$

$$where\, \rho = density\ \ of\ \ water, a\ =\ pressure\text{-}wave\ \ velocity,\ \wedge \text{¿} \Delta V\ =\ change\ \in\ flow\ \ velocity.$$

Consideration of Hydraulic Transients in Hydropower Design

The phenomenon is considered during the hydraulic and mechanical design of the water-conveyance system in the following ways:

i. **Determine the maximum pressure rise and pressure drop** due to turbine load changes, sudden shutdown, starting and rejection of load.
ii. **Design the penstock** to safely withstand the maximum transient pressure in addition to the normal static and dynamic pressures.
iii. **Provide a surge tank** where required. It reduces pressure fluctuations and controls excessive water hammer in long headrace tunnels, pressure tunnels and penstocks.
iv. **Select suitable turbine governing characteristics** so that wicket gates/nozzles do not close or open too rapidly.
v. **Determine safe valve and gate closing time.** Gradual closure reduces the magnitude of water hammer.
vi. **Check minimum pressure conditions** to prevent excessive pressure reduction, separation of water column and possible cavitation.
vii. **Analyze pressure waves** in the headrace, surge tank and penstock using transient-flow analysis for different operating conditions.
viii. **Provide pressure-relief/protection devices** such as appropriate valves or other control arrangements where necessary.
ix. **Check structural stability** of the penstock, anchors, bends, supports and other hydraulic structures against transient forces.
x. **Consider emergency conditions** such as sudden load rejection, turbine shutdown, power failure and rapid valve closure.

Given:

Internal diameter of penstock:

$$D = 2\,m = 2000\,mm$$

Statistical head:

$$H_s = 100\,m$$

Additional dynamic head $= 20\,\%$ of statistical head

Therefore, total design head is:

$$H = 100 + 0.20(100) = 120\,m$$

Allowable stress of material:

$$\sigma = 120\,N / mm^2$$

Joint efficiency:

$$\eta = 90\,\% = 0.90$$

For a thin cylindrical penstock, the hoop stress equation is:

$$\sigma = \frac{pD}{2t\eta}$$

$$where \quad p = \rho g H .$$

$$p = 1000 \times 9.81 \times 120$$

$$p = 1{,}177{,}200\,N / m^2$$

$$Converting \text{ ¿ } N / mm^2 :$$

$$p = \frac{1{,}177{,}200}{10^6} = 1.1772\,N / mm^2$$

$$t = \frac{pD}{2\sigma\eta}$$

$$t = \frac{1.1772 \;\; x \;\; 2000}{2 \;\; x \;\; 120 \;\; x \;\; 0.90}$$

$$t \approx 10.9\,mm$$

Hence, the theoretical wall thickness of the penstock is approximately 11 mm.

5. **Write the major parameters to decide the type of dam for storage projects. What major load you consider while designing the concrete gravity dam?**

Answer:

The major parameters to decide the type of dam for storage projects are as follows:

i. **Topography and valley shape**: A narrow, steep-sided valley generally favors an arch dam, while a wide valley may favor an earth or rock-fill dam.

ii. **Geological and foundation conditions**: The strength, permeability, stability and bearing capacity of the foundation are important. Strong rock foundations are suitable for concrete dams.

iii. **Availability of construction materials**: Local availability of earth, rock, aggregates and cement affects the choice and economy of the dam.

iv. **Height of dam and reservoir size**: Very high dams may require different structural solutions compared with low or medium-height dams.

v. **Hydrological conditions**: Design flood, spillway requirements, river discharge and sediment load influence dam selection.

vi. **Seismic conditions**: Earthquake intensity and seismicity of the region are especially important in Nepal.

vii. **Climate and environmental conditions**: Rainfall, temperature, construction season and environmental impacts should be considered.

viii. **Cost and construction time**: The selected type should be technically feasible and economically justified.

ix. **Availability of construction equipment and technology**: Local skills, machinery and construction techniques also influence the selection of the type of dam.

x. **Reservoir sedimentation**: High sediment yield may affect the type, height and arrangement of the dam and associated outlets.

The major loads that can be considered while designing the concrete gravity dam are:

i. **Dead load/self-weight of the dam**: Weight of the concrete body, which provides the main resisting force against sliding and overturning.

ii. **Water pressure (hydrostatic pressure)**: Pressure exerted by the reservoir water on the upstream face.

iii. **Uplift pressure**: Upward pressure acting at the base of the dam due to water seepage through the foundation and joints.

iv. **Earthquake/seismic forces**: Inertia forces due to earthquake acceleration and hydrodynamic water pressure caused by seismic motion.

v. **Temperature effects**: Stresses caused by variation in concrete temperature and thermal expansion/contraction.

vi. **Wind pressure**: Generally small but may be considered where applicable, particularly for exposed dam components.

6. **a. Define the terms load factor, capacity factor and utilization factor and relate them.**
b. Why is it necessary to predict future load demand. What are the methods of load forecasting?

Answer:

a. These are important performance indicators used to evaluate the operation of a hydropower plant.

i. **Load Factor**

Load factor is the ratio of the average load supplied by a power plant to its maximum demand (peak load) during a given period.

$$\text{Load Factor} = \frac{\text{Average Load}}{\text{Maximum Demand}}$$

It can also be written as:

$$Load\ Factor = \frac{Energy\ generated \vee supplied}{\text{Maximum Demand} \times \text{Time}}$$

A higher load factor indicates more uniform use of the plant.

ii. **Capacity Factor**

Capacity factor is the ratio of the actual energy generated by a plant during a given period to the energy that would have been generated if the plant operated continuously at its installed capacity.

$$Capacity\ Factor = \frac{Actual\ \ energy\ \ generated}{Maximum\ \ demand\ \ x\ \ time}$$

$$For\ \ a\ \ year:$$

$$\text{Capacity Factor} = \frac{E_{\text{actual}}}{P_{\text{installed}} \times 8760}$$

iii. **Utilization Factor**

Utilization factor is the ratio of the maximum demand or maximum load on the plant to its installed capacity.

$$Utilization\ Factor = \frac{Maximum\ \ demand}{Installed\ \ capacity}$$

It indicates how much of the installed generating capacity is actually utilized at peak demand.

Let:

$$P_{\text{avg}} = average\ \ load$$

$$P_{\text{max}} = maximum\ \ demand\ \ installed$$

$$P_{\text{installed}} = installed\ \ capacity$$

Then:

$$Load\ Factor = \frac{P_{\text{avg}}}{P_{max}}$$

$$\text{Utilization Factor} = \frac{P_{max}}{P_{\text{installed}}}$$

$$\text{Capacity Factor} = \frac{P_{\text{avg}}}{P_{\text{installed}}}$$

$$Therefore,$$

$$\text{Capacity Factor} = \text{Load Factor} \times \text{Utilization Factor}$$

or

$$CF = LF \times UF$$

b. Future load demand forecasting is necessary for planning, designing and operating a power system economically and reliably. It is necessary to predict the future demand because of the following reasons:

i. To determine the future generating capacity required to meet increasing electricity demand.
ii. To determine the required installed capacity, number and size of generating units, and the type of project required.
iii. To plan future transmission lines, substations and distribution networks.
iv. To estimate future investment requirements, revenue and economic feasibility of power projects.
v. To maintain adequate generation capacity and avoid shortages, load shedding and system overloading.
vi. In hydropower systems, forecasting helps in proper planning of water resources and reservoir operation.

vii. It helps determine how much power should be generated at different times and enables economical operation of generating units.

viii. It provides a basis for deciding when and how much additional generating capacity should be added.

The methods of load forecasting are listed below:

i. **Trend or extrapolation method**
Past electricity consumption and demand trends are analyzed and extrapolated to estimate future demand. It is simple and commonly used for long-term forecasting.

ii. **Econometric method**
Mathematical relationships are established between electricity demand and factors such as population, income, GDP, electricity price, industrial growth and urbanization.

iii. **Time-series method**
Historical load data are analyzed to identify trends, seasonal variations and other patterns. Methods such as regression and moving averages may be used.

iv. **Simulation method**
A model of the power system and consumer behavior is developed to simulate future demand under different conditions.

v. **Based on forecasting period**
Load forecasting is also classified as:

Short-term load forecasting: Hours to a few days ahead; mainly used for system operation.

Medium-term load forecasting: Weeks to several months or a few years; used for maintenance, fuel/resource planning and operational planning.

Long-term load forecasting: Several years to decades ahead; mainly used for generation, transmission and hydropower project planning.

7. **a. Write three financial indicators to decide financial viability of any hydropower project?**
b. Determine the discounted pay back period of a project that requires an investment of 100 million US $. Take minimum attractive rate of return as 15 %

Answer:

a. The three commonly used financial indicators are:

i. **Net Present Value (NPV):**
It is the present value of all future cash inflows minus the present value of cash outflows. A project is financially viable when NPV > 0.

ii. **Internal Rate of Return (IRR):**
It is the discount rate at which the NPV of the project becomes zero. The project is generally considered viable when IRR is greater than the required rate of return or cost of capital.

iii. **Benefit–Cost Ratio (B/C Ratio):**
It is the ratio of the present value of benefits to the present value of costs.

$$B/C = \frac{\text{Present Value of Benefits}}{\text{Present Value of Costs}}$$

A project is financially viable when B/C > 1.

Thus, the three indicators are: NPV, IRR and Benefit–Cost Ratio.

b. Initial investment:

$$I_0 = 100 \text{ million US\$}$$

$$Minimum\ \ attractive\ \ rate\ \ of\ \ return:$$

$$i = 15\,\%$$

Assume annual net cash inflow = US$30 million.

The discounted cash flow is:

$$DCF_t = \frac{CF}{(1+i)^t}$$

Year	Cash flow (million US$)	Discount factor @ 15%	Discounted cash flow	Cumulative DCF
1	30	0.8696	26.09	26.09
2	30	0.7561	22.68	48.77
3	30	0.6575	19.72	68.49
4	30	0.5718	17.15	85.64
5	30	0.4972	14.91	100.55

The initial investment of US$100 million is recovered during the 5th year.

The amount remaining after 4 years is:

$$100 - 85.64 = 14.36\ \ \text{million}$$

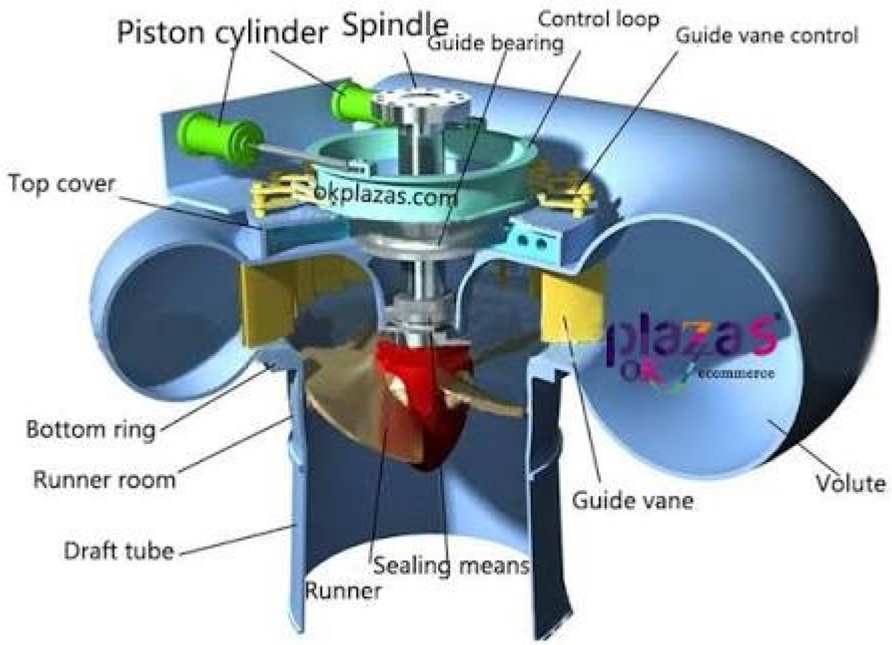
$$Fraction\ \ of\ \ the\ \ 5th\ \ year\ \ required:$$

$$\frac{14.36}{14.91} = 0.963$$

$$Therefore,$$

$$\text{Discounted Payback Period} = 4 + 0.963$$
$$\text{DPP} \approx 4.96\ \ \text{years}$$

8. **a. Why do you require the governor in a hydropower plant?**
b. What precautions will you take at blasting site if you are responsible for supervision of work?
Answer:
a. A governor is an automatic control device used to regulate the speed of the turbine by controlling the quantity of water supplied to it.



The governor is required in a hydropower plant for following purposes:

i. **Maintains constant turbine speed:** It maintains the turbine speed nearly constant despite changes in load.
ii. **Maintains constant frequency:** Since turbine speed determines generator frequency, the governor helps maintain the required electrical frequency, generally 50 Hz in Nepal.
iii. **Controls water flow:** It automatically increases or decreases the opening of the turbine guide vanes/nozzles according to the load requirement.
iv. **Responds to load changes:** When electrical load increases, the governor increases water flow to the turbine. When load decreases, it reduces water flow.
v. **Prevents excessive speed:** During sudden reduction or loss of load, it prevents the turbine from accelerating to dangerously high speeds.
vi. **Improves system stability:** It helps maintain stable operation of the turbine-generator unit and facilitates proper sharing of load when generators operate in parallel.

In hydropower projects, explosives may be required for tunnel excavation, road construction, quarrying, and other rock excavation works. Explosives used in hydropower and major infrastructure projects are strictly controlled. The Nepal Army has an important role in the security, controlled blasting and disposal of explosives, and hydropower projects coordinate with trained/authorized Army personnel for such operations. Proper storage, transportation, handling, blasting and disposal procedures are essential because explosives can cause fatal accidents.

Recent examples: On 10 August 2026, an explosion occurred at an Army post in Rasuwa during the disposal of expired explosives associated with hydropower/road blasting. One person was killed and six were injured, including two Army personnel. Another recent example is the Upper Trishuli 3A project in Nuwakot, where Nepal Army engineers carried out controlled blasting during rescue operations to clear blocked tunnel sections after the August 2026 floods.

If I am responsible for supervising blasting work, I would take the following precautions:

i. **Proper planning and authorization:** Ensure that blasting is carried out only by qualified and licensed personnel and according to the approved blasting plan and applicable regulations.
ii. **Inspection of the site:** Examine the site, surrounding structures, roads, utilities, tunnels, slopes and nearby settlements before blasting.
iii. **Safe storage of explosives:** Store explosives and detonators separately in approved, secure and clearly marked magazines away from unauthorized persons and sources of heat.
iv. **Controlled transportation:** Transport explosives carefully in approved containers/vehicles and keep them away from ignition sources and unnecessary personnel.

v. **Controlled access:** Establish a clearly marked danger/exclusion zone around the blasting area and prevent entry of unauthorized persons.

vi. **Warning signals:** Give adequate warning using sirens, flags, notices or other agreed signals before charging, firing and after completion of blasting.

vii. **Evacuation:** Ensure that all workers, visitors and nearby people are moved to a safe location before firing.

viii. **Protection from flying fragments:** Provide suitable blasting mats, screens or other protective measures where there is a risk of flyrock.

ix. **Weather and visibility:** Avoid blasting during unsafe weather conditions, particularly when lightning or other conditions could create additional hazards.

x. **Check before firing:** Confirm that all workers and equipment are clear, access roads are controlled, and the firing area has been inspected before initiating the blast.

xi. **Misfire precautions:** If a misfire occurs, do not approach or disturb the blast hole immediately. Isolate the area and allow only authorized blasting personnel to deal with it according to the approved procedure.

xii. **Post-blast inspection:** After firing, keep the area restricted until it has been declared safe. Check for misfires, unstable rock, loose materials, fumes and other hazards.

9. The mean monthly flow of typical Nepalese river is as follows:

Months	**Jan**	**Feb**	**Mar**	**Apr**	**May**	**Jun**	**Jul**	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**
Q(cumecs)	**8**	**7**	**6**	**7**	**10**	**38**	**90**	**100**	**60**	**30**	**16**	**10**

Considering the overall efficiency as 0.8 net head as 200m and minimum equivalent release as 10% of minimum monthly flow determine:

a. Total potential (power and energy) in each months considering no spilling is allowed.

b. If the power plant will run only for 6 months what will be design discharge and installed capacity of the plant?

Answer:

Given:

$$\eta = 0.8,$$

$$H = 200\, m$$

$$Minimum\ \ monthly\ \ flow\ \ is\ \ Q_{min} = 6\, m^3/s$$

$$minimum\ \ equivalent\ \ release\ \ is\ \ Q_r = 10\,\% \times 6 = 0.6\, m^3/s$$

$$Since\ \ no\ \ spilling\ \ is\ \ allowed,\ the\ \ discharge\ \ available\ \ for\ \ power\ \ generation \in each\ \ month\ \ is$$

$$Q_g = Q - Q_r = Q - 0.6$$

$$The\ \ power\ \ generated\ \ is$$

$$P = \eta \rho g Q_g H$$

$$¿,\ \in\ kW,$$

$$P = 9.81\, \eta Q_g H$$

$$Thus,$$

$$P = 9.81(0.8)(200)(Q - 0.6)$$

$$P = 1569.6(Q - 0.6)\, kW$$

(a) Monthly power and energy potential

Using the above equation:

Month	**Q (m^3/s)**	**Release (m^3/s)**	**Qg (m^3/s)**	**Power (MW)**	**Energy (GWh)**
Jan	8	0.6	7.4	11.62	8.642

Feb	7	0.6	6.4	10.05	6.751
Mar	6	0.6	5.4	8.48	6.306
Apr	7	0.6	6.4	10.05	7.233
May	10	0.6	9.4	14.75	10.977
Jun	38	0.6	37.4	58.70	42.266
Jul	90	0.6	89.4	140.32	104.400
Aug	100	0.6	99.4	156.02	116.078
Sept	60	0.6	59.4	93.23	67.129
Oct	30	0.6	29.4	46.15	34.333
Nov	16	0.6	15.4	24.17	17.404
Dec	10	0.6	9.4	14.75	10.977

The total annual energy potential is approximately

$$E_{\text{annual}} = 432.49\,GWh$$

Example calculation for August

For August,

$$Q_g = 100 - 0.6 = 99.4\,m^3/s$$

Therefore,

$$P = 9.81(0.8)(99.4)(200)$$

$$P = 156.02\,MW$$

For 31 days,

$$E = P \times 24 \times 31$$

$$E = 156.02 \times 24 \times 31$$

$$E = 116.08\,GWh$$

(b) Plant operating for only 6 months

The six months having the highest river discharge are:

$$May\,,June\,,July\,,August\,,September \wedge October$$

Their total discharge is

$$10 + 38 + 90 + 100 + 60 + 30 = 328\,m^3/s$$

Hence the average discharge over these six months is

$$Q_{\text{avg}} = \frac{328}{6}$$

$$Q_{\text{avg}} = 54.67\,m^3/s$$

After maintaining the minimum release:

$$Q_d = 54.67 - 0.6$$

$$Q_d = 54.07\,m^3/s$$

Therefore, the design discharge is approximately

$$Q_d \approx 54.1\, m^3/s$$

The installed capacity is

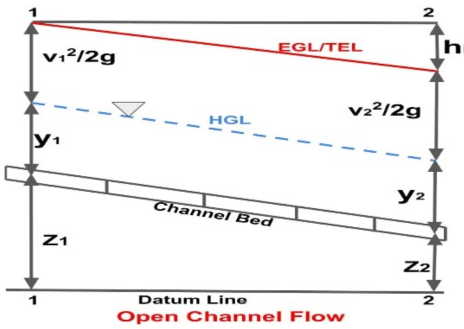
$$P_i = 9.81(0.8)(54.07)(200)$$

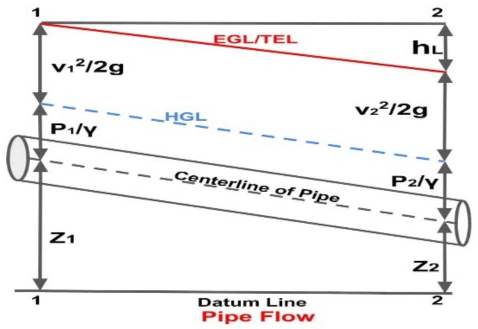
$$P_i \approx 85.0\, MW$$

10. **Differentiate between the pipe flow and open channel flow. A hydropower plant has a circular surge tank of diameter 15 m at the end of 2 km long tunnel having diameter 4m. Five penstocks each of diameter 1.525m and 400 m long are used Friction factor for tunnel is 0.018 and penstock is 0.03acoustic wave velocity in penstock is 1370 m/s. In steady state, head reservoir level is 450m with discharge at 25 m^3/s. Compute the water hammer pressure for sudden up closure, maximum upsurge and down surge and time of oscillation.**

Answer:

Fluid flow can occur either through the closed conduits under pressure or through open channels with a free surface. There are two types of flow differing in their governing principles and behavior. They are listed below:





Pipe flow	Open channel flow
Flow of liquid in a closed conduit running in a full flow is known as pipe flow.	Flow of liquid with a free surface exposed to atmosphere is known as open channel flow.
Pipe flow occurs due to the pressure difference between two points	Open channel flow is due to the channel slope, where water flows under gravity.
The flow area in the pipe flow is fixed (pipe cross-section remains constant)	Flow area in open channel flow is variable (depends on depth of flow)
Pipe flow depends on pressure gradient, pipe roughness, pipe material and its diameter.	Open channel flow depends on slope, depth, channel roughness and hydraulic radius.
Pressure energy is significant in pipe flow.	Potential energy (elevation head) dominates in open channel flow.
Pipe flow is determined by Darcy-Weisbach, Hazen-Williams equations	Open channel flow is determined by Manning's equation, Chezy's equation etc.
The flow in the pipe always runs full.	The flow in open channel may be uniform or non-uniform.
The parameters that control the pipe flow are: pressure, diameter, and length of pipe	The parameters that control the open channel flow area: Depth, slope, and cross-sectional shape
Examples of pipe flow: Water supply pipelines, sewer pipes (running full)	Examples of open channel flow: Rivers, canals, drainage channels

Thus, pipe flow is pressure-driven and occurs in closed conduits, whereas open channel flow is gravity-driven with a free surface, leading to fundamentally different analysis and design approaches.

Given

$$Tunnel\ \ length, L_t = 2000\,\mathrm{m}$$

$$Tunnel\ \ diameter\,, D_t = 4\,\text{m}$$

$$Tunnel\ \ friction\ \ factor\,, f_t = 0.018$$

$$Surge\ \ tank\ \ diameter\,, D_s = 15\,\text{m}$$

$$Number\ \ of\ \ \ penstocks\ = 5$$

$$Penstock\ \ diameter\,, D_p = 1.525\,\text{m}$$

$$Penstock\ \ length\,, L_p = 400\,\text{m}$$

$$Penstock\ \ friction\ \ factor\,, f_p = 0.03$$

$$Discharge\,, Q = 25\,\text{m}^3/\text{s}$$

$$Acoustic\ \ wave\ \ velocity\,, a = 1370\,\text{m}/\text{s}$$

$$Reservoir\ \ level = 450\,\text{m}$$

$$g = 9.81\,\text{m}/\text{s}^2$$

Velocity in each penstock

Area of one penstock:

$$A_p = \frac{\pi}{4}(1.525)^2 = 1.8265\,\text{m}^2$$

$$Since\ \ there\ \ are\ \ five\ \ penstocks\,,$$

$$Q_p = \frac{25}{5} = 5\,\text{m}^3/\text{s}$$

$$Therefore\,,$$

$$V_p = \frac{Q_p}{A_p} = \frac{5}{1.8265}$$
$$V_p = 2.737\,\text{m}/\text{s}$$

Water-hammer pressure for sudden closure

For sudden closure, the Joukowsky equation is

$$\Delta H = \frac{a V_p}{g}$$

$$Thus\,,$$

$$\Delta H = \frac{1370(2.737)}{9.81}$$

$$\Delta H = 382.3\, m\, of\ water$$

$$\Delta p = \rho g \, \Delta H$$

$$Taking \;\; = \rho = 1000 \, \mathrm{kg/m^3},$$

$$\Delta p = 1000(9.81)(382.3)$$
$$\Delta p \approx 3.75 \times 10^6 \, \mathrm{Pa}$$
$$\Delta p \approx 3.75 \, \mathrm{MPa}$$

Velocity in the tunnel

Tunnel area:

$$A_t = \frac{\pi}{4}(4)^2 = 12.566 \, \mathrm{m^2}$$

$$Therefore,$$

$$V_t = \frac{25}{12.566}$$
$$V_t = 1.989 \, \mathrm{m/s}$$

Maximum upsurge

For a surge tank, neglecting friction initially, the natural angular frequency of oscillation is

$$\omega = \sqrt{\frac{g A_t}{L_t A_s}}$$

$$where$$

$$A_s = \frac{\pi}{4}(15)^2 = 176.715 \, \mathrm{m^2}$$

$$Therefore,$$

$$\omega = \sqrt{\frac{9.81 \;\; x \;\; 12.566}{2000 \;\; x \;\; 176.715}}$$
$$\omega = 0.01868 \, rad/s$$

$$The \;\; ideal \;\; maximum \;\; upsurge \;\; is$$

$$Z_{max} = \frac{Q}{A_s \omega}$$

$$Z_{max} = \frac{25}{176.71 \;\; x \;\; 0.01868}$$

$$Z_{max} \approx 7.57 \, \mathrm{m}$$

Tunnel friction loss is:

$$h_f = f_t \frac{L_t}{D_t} \frac{V_t^2}{2g}$$
$$h_f = 0.018 \frac{2000}{4} \frac{(1.989)^2}{2(9.81)}$$

$$h_f \approx 0.91\,\text{m}$$

$$Including\ \ this\ \ frictional\ \ damping\,,\ \ the\ \ maximum\ \ upsurge\ \ is\ \ approximately$$

$$Z_{max\,,up} \approx 6.58\,\text{m}$$

$$Thus\,,\ \ the\ \ maximum\ \ water\ \ level\ \in\ the\ \ surge\ \ tank\ \ is\ \ approximately$$

$$450 + 6.58$$

$$H_{max\,,up} \approx 456.6\,\text{m}$$

With frictional damping, the first downsurge is smaller than the upsurge.

Solving the surge-tank oscillation including tunnel friction gives approximately

$$Z_{max\,,down} \approx 5.14\,\text{m}$$

$$Therefore\,,\ \ the\ \ minimum\ \ water\ \ level\ \ is$$

$$450 - 5.14450 - 5.14$$

$$H_{min} \approx 444.86\,\text{m}$$

Time of oscillation

$$The\ \ theoretical\ \ period\ \ of\ \ oscillation\ \ is$$

$$T = 2\pi\sqrt{\frac{L_t A_s}{g A_t}}$$

$$Substituting\,,$$

$$T = 2\pi\sqrt{\frac{2000\ \ x\ \ 176.715}{9.81\ \ x\ \ 12.566}}$$

or

$$T \approx 5.61\,\text{min}$$

The first maximum upsurge occurs approx. after one-quarter of the ideal oscillation period:

$$t_{up} \approx \frac{T}{4}$$

$$t_{up} \approx 84.1\,\text{s}$$

11. **List out the major activities varied out to prepare the feasibility study of hydropower. What is the purpose of subsurface exploration? Explain in brief how it is conducted?**

Answer:

The major activities involved in preparing a feasibility study of a hydropower project are:

i. **Collection and review of existing information**
Collect available topographical maps, geological maps, hydrological records, meteorological data, previous studies, cadastral information, and other relevant reports.

ii. **Reconnaissance survey and site investigation**
Visit the project area and identify possible locations for the dam/intake, waterway, powerhouse, access roads, transmission line, and other structures.

iii. **Topographical survey**
Carry out detailed topographical surveys to establish the river profile, contours, project layout, reservoir area, waterways, powerhouse site, and access routes.

iv. **Hydrological study**
Analyze rainfall, river discharge, flood, sediment load, low-flow conditions, and flow-duration curves to determine the dependable and design discharges.

v. **Geological and geotechnical investigation**
Study the geology, rock quality, faults, joints, landslides, foundation conditions, tunnelling conditions, and seismic characteristics of the project area.

vi. **Hydraulic and energy assessment**
Determine the available head, design discharge, installed capacity, annual energy generation, firm energy, and plant utilization factor.

vii. **Selection of project alternatives**
Examine alternatives for project type, dam/intake location, waterway alignment, powerhouse location, turbine type, and installed capacity, and select the most suitable option.

viii. **Preliminary design of major structures**
Prepare preliminary designs and layouts of the dam/intake, desanding basin, headrace tunnel/canal, surge tank, penstock, powerhouse, tailrace, access roads, and other hydraulic structures.

ix. **Environmental and social study**
Identify environmental impacts, land requirements, affected households, resettlement issues, ecological impacts, and propose mitigation and enhancement measures.

x. **Power evacuation and transmission study**
Study the nearest grid connection point, transmission line route, substation requirements, power evacuation capacity, and associated costs.

xi. **Construction planning and scheduling**
Estimate construction methods, construction period, access requirements, major construction facilities, manpower, equipment, and implementation schedule.

xii. **Cost estimation**
Prepare estimates of civil works, electromechanical equipment, hydromechanical equipment, transmission system, land acquisition, environmental mitigation, engineering, and other project costs.

xiii. **Economic and financial analysis**
Determine project benefits, revenue, financial viability, economic internal rate of return, benefit-cost ratio, payback period, sensitivity to cost and energy changes, etc.

xiv. **Risk and uncertainty assessment**
Assess risks related to hydrology, geology, construction, cost escalation, power generation, financing, environmental issues, and natural disasters.

xv. **Preparation of feasibility report**
Compile all technical, environmental, social, financial, and economic findings and recommend whether the project is technically feasible, economically viable, environmentally acceptable, and financially feasible.

Subsurface exploration is the investigation of the soil and rock conditions below the ground surface to obtain information necessary for the safe and economical design and construction of foundations and other civil engineering structures.
The main purpose of subsurface exploration are as follows:

i. To determine the nature, thickness and sequence of soil and rock strata.

ii. To determine the depth of bedrock and its engineering properties.
iii. To evaluate soil properties such as shear strength, compressibility, permeability and density.
iv. To determine the bearing capacity and settlement characteristics of the foundation soil.
v. To locate the groundwater table and study its seasonal variation.
vi. To identify weak layers, faults, cavities, expansive soils, landslides and other unfavorable conditions.
vii. To obtain information required for selecting the type, depth and size of foundation.
viii. To assess construction problems such as excavation stability, dewatering and tunnelling conditions.
ix. To obtain representative disturbed and undisturbed soil/rock samples for laboratory testing.

The exploration is generally carried out in the following steps:

i. **Desk study:**
Available geological maps, topographic maps, aerial photographs, previous investigation reports and other site information are reviewed.

ii. **Site reconnaissance:**
The site is visually inspected to identify surface geology, drainage, cracks, landslides, rock outcrops, groundwater seepage and other features.

iii. **Planning of investigation:**
The number, location and depth of boreholes or trial pits are decided according to the type and importance of the proposed structure.

iv. **Field exploration:**
Subsurface conditions are investigated using methods such as trial and trench pits, boreholes, standard penetration text, cone penetration test and plate load test.

v. **Sampling:**
Soil and rock samples are collected from different depths. Undisturbed samples are obtained where required for strength and consolidation tests.

vi. **Groundwater observation:**
The groundwater level is measured in boreholes or observation wells.

vii. **Laboratory testing:**
Collected samples are tested for properties such as grain size, Atterberg limits, density, permeability, shear strength, consolidation and rock strength.

viii. **Interpretation and reporting:**
Field and laboratory results are analyzed to prepare soil/rock profiles, groundwater conditions and engineering parameters, and recommendations are made for foundation design and construction.